MAE 527: Final Project

Overview:

This crew module is designed for the Ariane 6 launch vehicle. Ariane Space has expressed future interest in creating a crew vehicle for this launch system, as none exists just yet. Our team is proposing an initial preliminary design for the program.

Launch Vehicle Background:

Launch Vehicle: Ariane 6

The Ariane 6 rocket is shown here with a two-booster configuration. We will be using the four-booster configuration which allows for more payload weight. The module will be sent to Low Earth Orbit in order to dock with the ISS. We will assume that CRV will account for a majority of the 200000 kg payload. The height from the fairing to the cryogenic clamp arm is 22.5m, estimated to be 1/3 of the 63m total height. The total thrust produced at launch will be 19370 kN. Ariane space launches from the European Spaceport in French Guiana, which experiences wind shear of up to 18.75 kN. The module radius will be 2.7m at its widest point to fit within the fairing [3] [4].

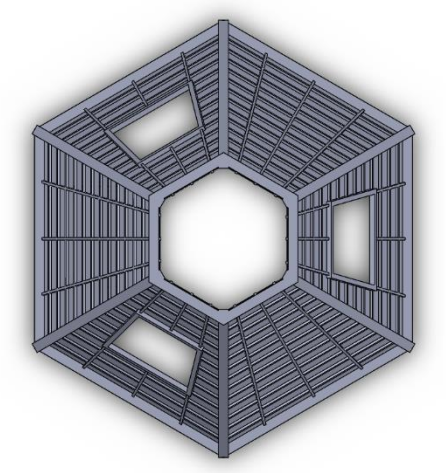
The module will also possess two windows and a hatch of equal size for radial symmetry, not typically seen in CRVs

https://www.esa.int/Enabling_Support/Space_Transportation/Launch_vehicles/Ariane_6_overview

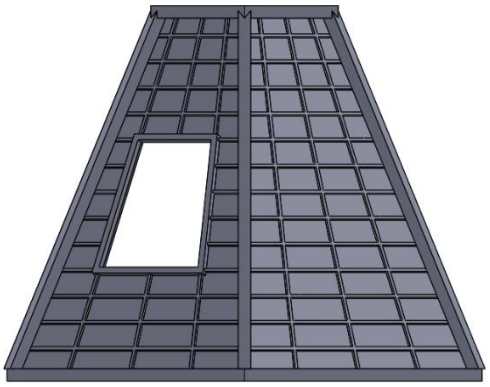
https://www.esa.int/Enabling_Support/Space_Transportation/Ariane/The_engines_of_Ariane_6

Module CAD:

Top:

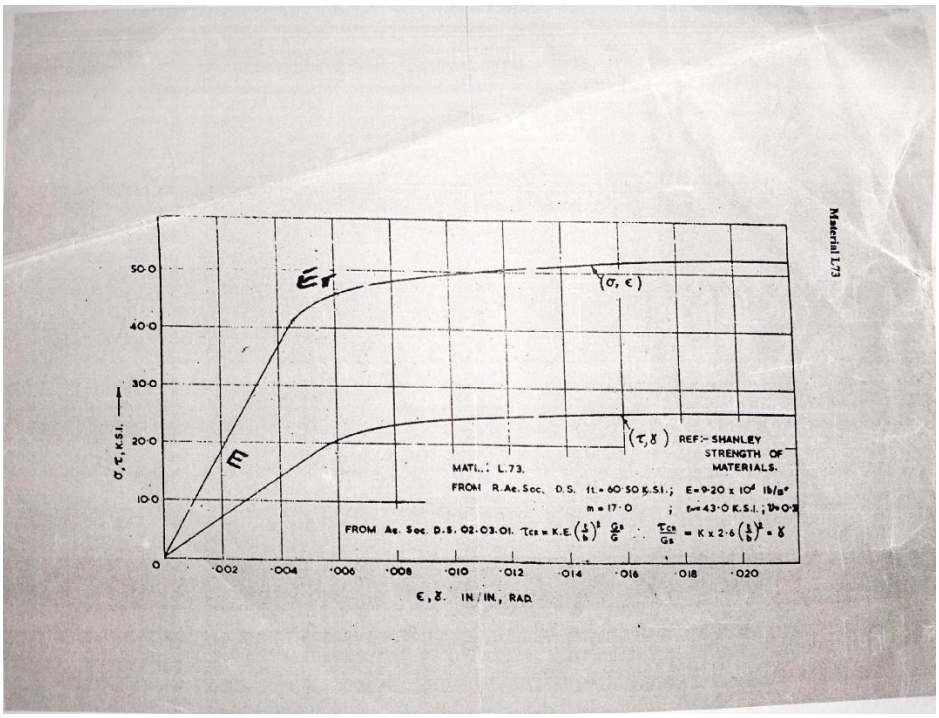


Left:



Isometric:

Material: L-73 Aluminum [1]



Compression Yield Point:

$\sigma \approx$ 41.4 ksi (2.8544e + 8 Pa), $\varepsilon \approx 0.0045$, $E \approx 9200\ ksi\ (6.34318e + 10\ Pa)$

Compression Proof Stress:

$\sigma_{proof} = 9200(0.0045)(1.02) = 42.2ksi\ (2.9096e + 8\ Pa)$

Shear Yield Point:

$\sigma \approx$ 21 ksi (1.448e + 8 Pa), $\varepsilon \approx 0.006$, $E \approx 3500\ ksi\ (2.4132e + 10\ Pa)$

Compression Proof Stress:

$\sigma_{proof} = 3500\ (0.006)(1.02) = 21.4\ ksi\ (1.4755e + 8\ Pa)$

<u>Displacement and Angular Displacement caused by wind shear upon launch:</u>

Background:

Wind Shear: Based on weather data gathered from days when the Ariane 6 launch vehicle launched showed that the maximum wind shear was approximately 1.875e4 (N) [2].

Parameter	Value	Units
Cylinder Height (Relative 1/3 Height from last support), H	22.5	m
Module Height, h	3	m
Wind Shear Force, D	18750	N
Upper Taper Radius, r1	0.65	m
Lower Taper Radius, r2	2.7	m
Young's Modulus, E	$6.34318e+10$	Pa
Shear Modulus, G	$2.4132e+10$	Pa
Shell Thickness, t	0.005	m
Module Mass, M	20000	kg

Example Calculation [1]:

$$d = d_{top\ of\ rocket-COG\ module} = \frac{H}{3}\left(1 + \frac{r1}{r2} + \left(\frac{r1}{r2}\right)^2\right) = 1.29$$

$$I = I_{Tapered\ Fustrum} = \frac{1}{3}M\left(r2^2 + r1^2 + \left(\frac{r2 \times r1}{3}\right)\right) = 5.5e4$$

$$Moment = D_{shear} \times H_{Relative\ Hieght\ of\ Rocket} = 421875\ Nm$$

$$G_{\delta\delta} = \left(\frac{\left(\frac{H}{r2}\right)^3}{3\pi Et} + \frac{\frac{H}{r2}}{\pi Gt}\right) = 2.1480e-07$$

$$G_{\delta\theta} = \frac{\left(\frac{H}{r2}\right)^2}{2\pi Et(r2)} = 1.2907e-08$$

$$G_{\theta\delta} = \frac{\left(\frac{H}{r2}\right)^2}{2\pi Et(r2)} = 1.2907e-08$$

$$G_{\theta\delta} = \frac{\left(\frac{H}{r2}\right)}{\pi Et(r2)^2} = 1.1473e-09$$

Matrix Assembly:

$$\boldsymbol{G} = \begin{bmatrix} G_{\delta\delta} & G_{\delta\theta} \\ G_{\theta\delta} & G_{\theta\delta} \end{bmatrix}$$

$$\boldsymbol{M} = \begin{bmatrix} M & 0 \\ 0 & I + \frac{1}{2}M(d)^2 \end{bmatrix}$$

$$\boldsymbol{M}\begin{bmatrix}\ddot{\delta}\\ \ddot{\theta}\end{bmatrix} + \boldsymbol{G}^{-1}\begin{bmatrix}\dot{\delta}\\ \dot{\theta}\end{bmatrix} + \begin{bmatrix}D_{shear}\\ Moment\end{bmatrix} = 0$$

$$\delta = 0.0013\ m$$

$$\theta = -0.0026\ m$$

Pressure Vessel Compliance:

Parameter	Value	Units
Young's Modulus, E	6.377×10^{10}	Pa
Shear Modulus, G	2.413×10^{10}	Pa
Poisson's Ratio, ν	0.3	–
Ambient Pressure, P	1.01314×10^{5}	Pa
Number of Stringers (per side), n_s	3	–
Stringer Cross-sectional Area, A_s	0.000625 (0.025x0.025)	m^2
Maximum Radius, r	2.7	m
Maximum Stringer Spacing, b	0.51	m
Maximum Flange Spacing, L	0.20	m
Shell Thickness, t	0.005	m
Skin Area, Askin	3.63	m^2
Maximum Axial Strain, ϵ_c	$\frac{1}{E}\left(\left(\frac{Pr}{t}\right) - \mu\left(\frac{Pr}{2t}\right)\right) = 7.29\times10^{-4}$	–
Radial Expansion, v	$\epsilon_c \times r = 0.0020$	m
Spacecraft Weight, M_s	1.96×10^{5}	N
Rocket Launch Force, F_{Rocket}	1.937×10^{7}	N

Example Calculation [1]:

$$k = k_{(stringer\ area)/(skin\ area)} = \frac{(\mathrm{n_s}\times\mathrm{A_s})}{Askin} = 5.1796e-04$$

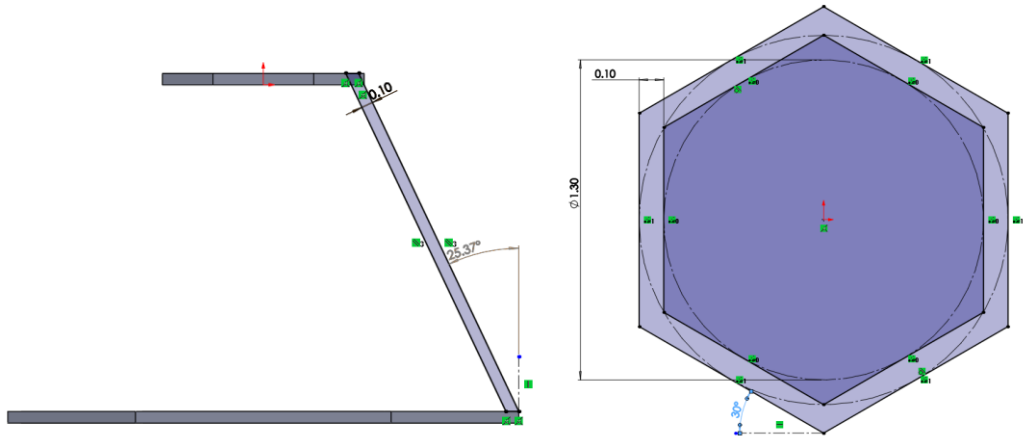
Skin stress in longitudinal direction: $fm = \frac{Pr}{2t}\left(\frac{1+2\nu k}{1+k}\right) = 2.7352e+07$

Stringer stress in longitudinal direction: $fls = \frac{Pr}{2t}\left(\frac{1-2\nu}{1+k}\right) = 1.0937e+07$

Skin Stress around the circumferential direction (Hoop Stress): $fs = \frac{Ev}{r} + \mu\frac{Pr}{t} = 3.8301e+07$

Applied Shear Load to from at each stringer location: $V = bL(0.85P - \left(\frac{Et}{r^2}\right)\nu) = -1.3296\mathrm{e}+06$

Compression Stress from skin:



$$fcomp = fm \times \sin(25.37°) + fs \times \sin(30°) = 3.0870e + 07$$

Critical Buckling Stress for curved panel (shear/compression):

Proof Stress Compression: $\sigma_{compression\ critical} = 2.76 \times 10^8\ Pa$

Proof Shear Compression: $\sigma_{Shear\ critical} = 1.35 \times 10^8\ Pa$

Critical Pressure:

$$P_{critical} = \frac{\sigma_{compression\ critical} \times t}{r} = 5.1193e + 05\ Pa; \text{ Comes from } (f_c = \frac{Pr}{t})$$

Design Review:

$$Compresion\ Check = (\frac{fcomp}{\sigma_{compression\ critical}})^2 - \frac{P}{P_{critical}} = 0.1855; \text{ Should be less then one so pass.}$$

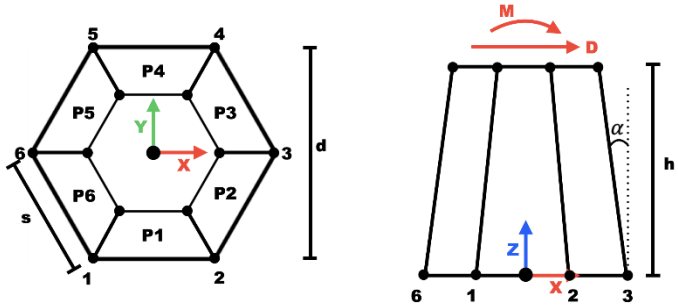
$$Shear\ Check = (\frac{V}{\sigma_{shear\ critical}})^2 - \frac{P}{P_{critical}} = 0.1978; \text{ Should be less then one so pass.}$$

Tapered Hexagon Box:

The shear force in the x-direction, D_x, is simply shear force D . The sear force in the y-direction, D_y, is zero. The moment in the x-direction, M_x, is zero and the moment in the y-direction, M_y, is governed by the equation below.

$$M_y = M + D(h - z)$$

Where M is the applied moment, h is the height of the CRV, and z is the length from the CRV's fixed end attached to the launch vehicle following the nomenclature in the figure below. The figure below also displays the panel numbers such as P1, P2, P3, P4, P5, and P6.



The side length of the CRV, s , is a function of z due to the changing cross-sectional area of the CRV as the result of its taper. The equation to determine the side length at any height of the CRV is below.

$$s_c = \frac{d - 2ztan\alpha}{\tan 60}$$

Nominal stress due to bending can be determined using the equation below.

$$\sigma_z = \frac{M_y x}{I_{yy}}$$

Using the figure above, the area moments of inertia in the yy-direction is determined to be

$$I_{yy} = 4A_B\left(\frac{1}{2}s_c\right)^2 + 2A_B s_c^2 = 3A_B s_c^2$$

Where A_B is the cross-sectional area of the boom. For booms 3 and 6, $x = s_c$ and for booms 1, 2, 4, and 5, $x = \frac{s_c}{2}$.

Therefore, the nominal stress in the z-direction caused by the bending from the moment and shear force is determined by the equation below for booms 3 and 6.

$$\sigma_z = \frac{M_y}{3A_B s_c}$$

The nominal stress in the z-direction for booms 1, 2, 4, and 5 are determined by the equation below.

$$\sigma_z = \frac{M_y}{6A_B s_c}$$

To determine the nominal stresses in the x, σ_x, and y, σ_y, directions, the following equations are used.

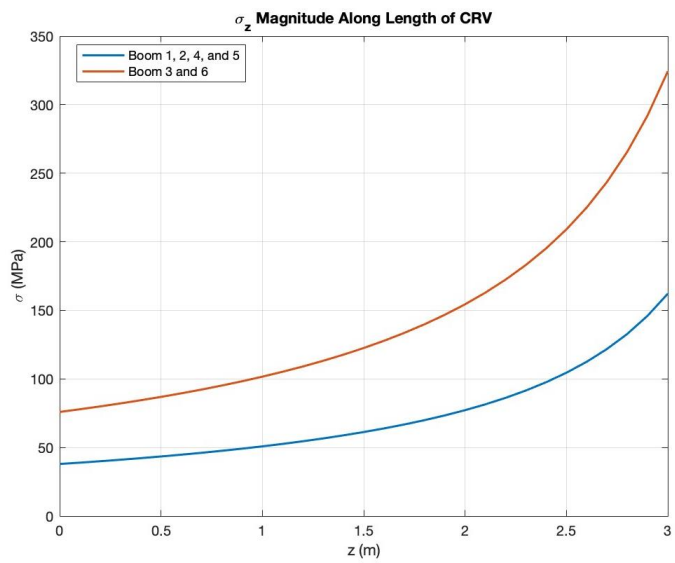
$$\sigma_x = \sigma_z \frac{\delta x}{\delta z}$$

$$\sigma_y = \sigma_z \frac{\delta y}{\delta z}$$

Where $\frac{\delta x}{\delta z}$ is the change in boom location in the x-direction over the length wise direction and $\frac{\delta y}{\delta z}$ is the change in boom location in the y-direction over the length wise direction. Then, the axial stress, σ , is determined using the equation below.

$$\sigma = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2}$$

The follow results are under the assumptions that the external loads include a shear force of 18.75 kN and a moment of 421.875 kN-m as well as CRV geometry of a base diameter of 3.7 m, height of 3 m, taper angle of 25.37°, and a boom cross-sectional area of 0.01 m^2. The figure below shows the magnitude of the axial stress in each boom. The taper causes the axial stress in the booms to increase towards the top of the CRV.



Shear flow originates from the equation below.

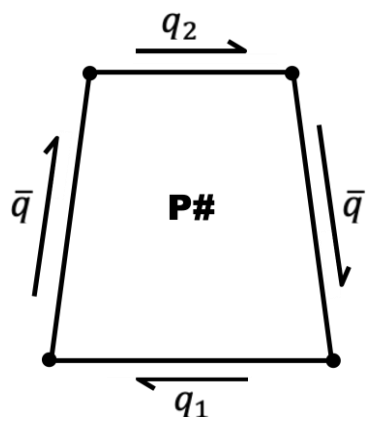
$$q = -\left(\frac{D_x I_{xx} - D_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2}\right)\left(\int_0^s txds + \sum_{r=1}^{n} A_{B,r}\, x_r\right) - \left(\frac{D_y I_{yy} - D_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2}\right)\left(\int_0^s tyds + \sum_{r=1}^{n} A_{B,r}\, y_r\right)$$

$D_y = 0$ and due to symmetry, $I_{xy} = 0$ and all boom cross-sectional areas are equal therefore, the equation above simplifies to the equation below.

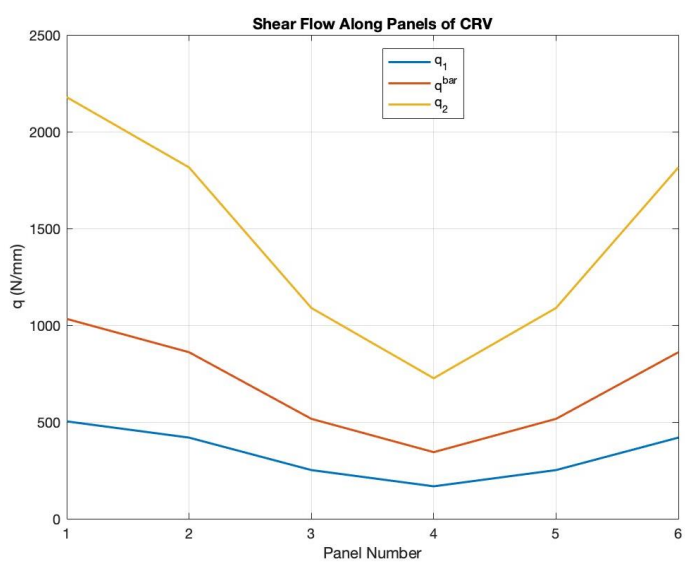
$$q = \frac{-D_{x,w}}{I_{yy}} A_B \sum_{r=1}^{n} (x_r\,) + q_0$$

Where $D_{x,w} = D_x - \sum_{r=1}^{m} P_{z,r} \frac{\delta x_r}{\delta z}$ which $P_{z,r}$ is the force on the boom in the z-direction.

The shear forces were first solved as an open section and then using the summation of moments about the center to the hexagonal shape of the CRV, the constant of integration, q_0, was determined. Following the convention shown in the figure below, table below said figure shows the shear flows of each panel.



Panel Number	q_1 (N/mm)	$\bar{q}$ (N/mm)	q_2 (N/mm)
1	503.278	1032.66	2178.34
2	419.398	860.554	1815.28
3	251.639	516.332	1089.17
4	167.759	344.222	726.112
5	251.639	516.332	1089.17
6	419.398	860.554	1815.28



Cutout Compliance:

Cutout area calculation

Parameter	Value	Units
Area of frames/stringers A	6.25*10^-4	m^2
Force on each stringer/frame P	6.8*10^3	N
Distance between stringers b	0.510	m
Length of stringers L	0.200	m
Young's Modulus, E	6.377×10^{10}	Pa
Shear Modulus, G	2.413×10^{10}	Pa
Shell Thickness, t	0.005	m

$P_1 = \frac{3P}{2}$ 3 stringers, symmetric distribution

$$A_L = \frac{A}{2}$$

$A_T = A + \frac{A}{2} = \frac{3A}{2}$ = 937.5mm

$$K = \sqrt{\frac{Gt}{Eb}\left(\frac{2}{A} + \frac{1}{A}\right)} = \sqrt{\frac{2.413 \cdot 10^{10} Pa \cdot 0.005m}{63877 \cdot 10^{10} Pa \cdot 0.510m}\left(\frac{2}{0.025m^2} + \frac{1}{0.025m^2}\right)} = 4.2313m$$

$$\sigma_F = \frac{P}{A_T}\left(1 + \frac{A_L}{A} \cdot \frac{\sinh(KL)}{\sinh(KL)}\right) = \frac{3P}{2A_T} \cdot (1.5) = \frac{3 \cdot 6.8 \cdot 10^3 N}{2 \cdot 937.5mm^2}(1.5) = 16.32MPa$$

$$A_c = \frac{P_1}{\sigma_F} = \frac{3P}{2\sigma_F} = \frac{3 \cdot 6.8 \cdot 10^3 N}{16.32MPa} = 625mm^2$$

For a square hatch frame, this would give a thickness of 25mm

Pressure Vessel with respect to cutout:

Parameter	Value	Units
Ambient Pressure, P	1.01314×10^5	Pa
Maximum Radius, r	2.7	m
Maximum Height cutout, H	1.10	m
Maximum Length cutout, L	0.914	m
Flange Thickness, a	0.05	m

Example Calculation [1]:

Pressure Vessel:

Distributed force along circumferential direction: $wc = P \times r = 2.7358e + 05$ Nm

Distributed force along longitudinal direction: $wl = P \times r = 1.3679e + 05$ Nm

Distribution Factor: $k = \frac{H}{(H+L)} = 0.5462$

Upper/Lower Moment: $M_{ul} = \frac{(wc \times L^2)}{12} = 1.9045e + 04\ Nm$

Front/Aft Moment: $M_{fa} = \frac{(wl \times H^2)}{12} = -1.3793e + 04\ Nm$

Shear Flow in upper front section: $q_{uf} = \frac{0.5(M_{ul} - k(M_{ul} + M_{fa}))}{a^2} = 3.2353e + 06\ N/m$

Stress in Section: $\sigma = q_{uf} \times a = 6.4706e + 07\ Pa < 2.76 \times 10^8\ Pa$ Proof Stress

Warping:

Initial Parameters

Parameter	**Value**	**Units**
Side Length, s	2.19	m
Wall Thicknesses, t_i	0.1	m
Enclosed Area, A_E	12.466	m^2
Radius to Wall Midpoint, r_t	1.8	m
Shear Modulus, G	2.29×10^{10}	Pa
Applied Torque, T	1500*	Nm

*1500Nm was taken as a mean estimate for torque, based on the Hubble Telescope's re-entry module [6]

Intermediate Calculations

- Torsional Constant J:

$J = (4 * A_E^2) / \Sigma(s/t_i)$ [5]
$\Sigma(s/t_i) = 5*(2.19/0.1)+(2.19 / 0.05) = 10 * 2.19 + 43.8 = 65.7$ m
$J = (4 * 12.466^2) / 65.7 = 621.47 / 65.7 = 9.462\ m^4$

- Twist per Unit Length θ':

$\theta' = T / (G * J)$
$\theta' = 50 / (2.29 \times 10^{10} \times 9.462) = 2.31 \times 10^{-10}$ rad/m

- Warping Constant K:

$K = \Sigma(1/t_i)/\Sigma(r_i)$ (derived from Saint-Venant's Equations)
$K = (5 * 1/0.1 + 1/0.05) / (6*1.8) = (10 + 20) / 10.8 = 2.78$

Warping Displacements

Node	**ϕ_i**	**$w_i = \phi_i * \theta'$ (m)**
A	0.000	0.00e+00
B	10.951	2.53e-09
C	21.902	5.06e-09
D	32.853	7.59e-09
E	65.728	1.52e-08
F	76.679	1.77e-08

Summary of Warping Calculations

Parameter	**Value**	**Units**
Side Length, s	2.19	m
Wall Thicknesses, t_i	0.1	m
Torsional Constant, J	9.462	m^4
Twist per Unit Length, θ'	2.31×10^{-10}	rad/m
Warping Constant, K	2.78	m^{-1}
Maximum Warping Displacement, w_max	1.77×10^{-8}	m

Example Calculation

$w_C = \phi_C \times \theta' = 21.902 \times 2.31 \times 10^{-10} = 5.06 \times 10^{-9}$ m

Bibliography:

[1] Bayandor, Javid. "MAE 527: Space Structures."

[2] The Weather Channel. "Europe Satellite Weather Map." *Weather.com*, The Weather Channel, https://weather.com/maps/satellite/northern-europe-weather-map. Accessed 19 May 2025.

[3] "Ariane 6 Overview." Esa.int, 2014, www.esa.int/Enabling_Support/Space_Transportation/Launch_vehicles/Ariane_6_overview.

[4] “The Engines of Ariane 6.” Esa.int, 2024, www.esa.int/Enabling_Support/Space_Transportation/Ariane/The_engines_of_Ariane_6.

[5] Megson, T.H.G. — *Aircraft Structures for Engineering Students- "Torsion of Thin-Walled Closed Sections*

[6] An updated re-entry analysis of the Hubble Space Telescope- Kyle Baker (1) , Eryn Culton (1) , Jonathan Lang (1), Zachary Lewis(1), Robert Perez-Alemany (1), Alexa Rizzo (1), Brendon Smeresky (1), Anthony Starks(1), Joshua Teneyck (1), Jennifer Rhatigan (jlrhatig@nps.edu) (2) and Marcello Romano

Appendix:

Displacement and Angular Displacement Calculations

Space Structures Displacement and Angle Calculation

```
clc; clear
% Given:
H=22.5; % Relative Hight of cylinder rocket (m)
h=3; % Height of spacecraft
D=1.875e4; % Wind Shear (N)
r_1=1.3/2; % Upper Taper Radius (m)
r_2=2.7; % Lower Taper Radius (m)
E=6.377e10; % Modulus (Pa)
G=3.63e3*6.895e+6; % Shear Modulus Pascals
t=0.005; % Thickness of cylinder (m)

% Calculated values for module
M_spacecraft = 20000; % mass of modulue (kg)
d= 3/3 * (1 + r_1/r_2 + (r_1/r_2)^2); % ditance from top of module to center of gravity (m)
I=(1/3) * M_spacecraft * (r_2^2 + r_1^2 + (r_2*r_1)/3); % Inertia of module (kg-m^2) (tapered
fulstrum)

%Input Shear and Moment Matrix
Moment=D*H; % moment from wind shear
Input=[D;Moment]; % Shear and moment matrix
```

G-Matrix

```
G_delta_delta = ((H/r_2)^3)/(3*pi*E*t)+(H/r_2)/(pi*G*t);
G_delta_theta = (H/r_2)^2/(2*pi*r_2*E*t);
G_theta_delta = (H/r_2)^2/(2*pi*r_2*E*t);
G_theta_theta = (H/r_2)/(pi*r_2^2*E*t);

G=[G_delta_delta G_delta_theta;G_theta_delta G_theta_theta];
Ginv=G^-1;
```

```
Dis_AnG_no_craft=G*Input; % no spacecraft displacment
```

M-Matrix

```
M=[M_spacecraft 0;0 I+d^2*M_spacecraft];

displacment_data=roots([M(1,1) Ginv(1) -Input(1)]);
Angle_data=roots([M(2,2) Ginv(2) -Input(2)]);

DisplacmentFinal=displacment_data(2)
AngleFinal=Angle_data(2)
```

```
DisplacmentFinal =

    0.0013


AngleFinal =

   -0.0026
```

Pressure Vessel Calculations:

Space Structures Pressure Vessel

```
clc; clear
% Given
E = 6.377e10; % Modulus (pa)
G = 2.4132e10; % Shear Modulus (Pa)
u = 0.3; % posions ratio
P = 101325; % ambient pressure at sea level (Pa)
ns = 3;  % number of stringers (#)
As = 0.025*0.025; % area of stringergs (m^2)
r=2.7; % largest radius of craft (m)
b=0.51; % largest distance between stringers (m)
L=0.20; % largest distance between fladges (m)
t=0.005; % wall thickness (m)
Askin =3.62; % Measured skin area
ec=(1/E)*((P*r)/t-(u*P*r)/(2*t));  % maxium strain with no 1/E*(fc-ufm)
v=ec*r; % expanded distance (m)
M_spacecraft = 20000*9.8; % mass of modulue (N)
frocket = 19370*1000; % rocket launch force (N)
```

solved variables

```
k = (ns*As)/Askin; % k: ratio of stringer area to skin area
fm = ((P*r)/(2*t))*((1+2*u*k)/(1+k)); % fm: skin stress in logitidial direction (N)
fls = ((P*r)/(2*t))*((1-2*u)/(1+k)) %fls: stringer stress in logituidal direction (N)
fc = ((E*v)/(r))-((u*P*r)/(2*t)); %fc: frame circumfretual stress (N)
fshear = (b*L)*((0.85*P)-((E*t)/r^2)*u); % shear force (N)
fcomp = fm*sind(25.37)+fc*sind(30); % total compressive force (N)
```

```
fls =

   1.0937e+07
```

Critical Points

Given

```
stress_compression_critical= (E)*(0.00425*1.02); % proof stress
stress_shear_critical= (G)*(0.0055*1.02); % proof stress
P_critical=stress_compression_critical*t/r; % fc=P*r/t
```

Review

```
Rp=P/P_critical;
Rc=fcomp/stress_compression_critical;
Rs=fshear/stress_shear_critical;
Compression_check=abs(Rc^2-Rp) % should less the 1
Shear_check=abs(Rs^2-Rp) % should less the 1
```

```
Compression_check =

    0.1855


Shear_check =

    0.1978
```

CUTOUT

```
hc = 1.1; % height cutout (m)
Lc = 0.914; % length cutout (m)
wc = P*r;
wl = P*r/2;
```

```
a=0.05;

k1 = hc/(Lc+hc); % disribtion factor
Mul = (wc*Lc^2)/12; % upper/lower moment
Mfa = (-wl*hc^2)/12; % front/aft moment

quf=(0.5*(Mul-k1*(Mul+Mfa)))/(a^2)

stress=quf/a
```

```
quf =

   3.2353e+06


stress =

   6.4706e+07
```

Warping Displacements

```
clear; clc;

% geometry
s = 2.19;  % side length [m]
t = [0.005, 0.005, 0.005, 0.005, 0.005, 0.005];  % uniform wall thickness [m]
A_E = (3 * sqrt(3) / 2) * s^2;  % area of regular hexagon
rt = 3.6 / 2;  % radial distance from center to wall midpoint [m]
rvec = rt * ones(1, 6);  % radius vector (same for all sides for now)

% material
G = 2.29e10;  % shear modulus [Pa]
T = 1500;       % applied torque [Nm];

J = (4 * A_E^2) / sum(s ./ t); % torsional constant J (thin-walled closed section)
theta_dot = T / (G * J); % twist per unit length

% K using closure condition
K = sum(1 ./ t) / sum(rvec);  % phi returns to original value after loop

% geometric warping function phi (unitless shape function)
phi = zeros(1, 6);
for i = 1:5
    dphi = s * (1 / t(i) - K * rvec(i));
    phi(i+1) = phi(i) + dphi;
end

% closure check (should return to original)
phi_A_check = phi(6) + s * (1 / t(6) - K * rvec(6));
```

```
% physical warping displacements w = phi * dtheta/dz
w_physical = phi * theta_dot;

% Output

fprintf('\nPhysical warping displacements w = phi * theta_dot:\n');
for i = 1:6
    fprintf('w_%c = %.6e m\n', 'A' + i - 1, w_physical(i));
end

fprintf('\nClosure check (phi_A again) = %.6e (should be = 0)\n', phi_A_check);
```

```
Physical warping displacements w = phi * theta_dot:
w_A = 0.000000e+00 m
w_B = 1.725183e-20 m
w_C = 3.450366e-20 m
w_D = 5.175550e-20 m
w_E = 6.900733e-20 m
w_F = 8.625916e-20 m
Closure check (phi_A again) = 3.734613e-13 (should be = 0)
```

Cutout Thickness

```
%Stringers
 %Assume constant Area, Thickness, and distance between the stringers. there
 %are 3 stringers and the assembly is symmetric.
 %assume a force P is applied to each stringer


A = 025*025; %Stringer area
 P = 6.8*10^3; %Force on each stringer
 G = 2.4132*10^10;
 E = 6.343*10^10;
 t = 5;
 b = 510; %distance between stringers
 AT = 3*A/2;
 K = sqrt(((G*t)/(b*E))*(3/A));
 L=200;


%Section 1: Case 1


SL1 = 0;
 SF1 = ((3*P)/(2*AT))*(1.5)
 TL= (P*K)/t*(A/(2*AT))*(sinh(K*L)/cosh(K*L))


AC = (3*P/2)/SF1
```

```
SF1 =

   16.3200


TL =

    1.3218


AC =

   625
```

Code for tapered box and shear flow

```
clc; clear;
```

Initialization

```
% External forces
D = 18750; % Shear force (N)
M = 4210875; % Moment [N m]


% Structure geometry
d = 3.7; % Diameter [m]
alpha = 25.37; % Taper angle [degrees]
A_B = 0.01; % Boom crossectional area (m^2)
H = 3; % Height (m)
t = 1.77*10^-4; % Thickness (m)

dz = 0.1; % Distance step
z = 0:dz:H; % Discritized length of structure

% Initializing vectors with zeros
sigma_Z1 = zeros(1,length(z)); % Stress in z-direction of boom 1
sigma_Z2 = zeros(1,length(z)); % Stress in z-direction of boom 2
sigma_Z3 = zeros(1,length(z)); % Stress in z-direction of boom 3
sigma_Z4 = zeros(1,length(z)); % Stress in z-direction of boom 4
sigma_Z5 = zeros(1,length(z)); % Stress in z-direction of boom 5
sigma_Z6 = zeros(1,length(z)); % Stress in z-direction of boom 6

P_Z1 = zeros(1,length(z)); % Stress in z-direction of boom 1
P_Z2 = zeros(1,length(z)); % Stress in z-direction of boom 2
P_Z3 = zeros(1,length(z)); % Stress in z-direction of boom 3
P_Z4 = zeros(1,length(z)); % Stress in z-direction of boom 4
P_Z5 = zeros(1,length(z)); % Stress in z-direction of boom 5
P_Z6 = zeros(1,length(z)); % Stress in z-direction of boom 6

sigma_2 = zeros(1,length(z)); % Axial stress of boom 2
sigma_3 = zeros(1,length(z)); % Axial stress of boom 3
```

Computation

```
for i = 1:1:length(z)
    M_y = M + D*(H-z(i));
    s_z = (d-2*z(i)*tand(alpha))/tand(60);
```

```
    sigma_Z1(i) = M_y/(6*A_B*s_z);
    sigma_Z2(i) = -M_y/(6*A_B*s_z);
    sigma_Z3(i) = -M_y/(3*A_B*s_z);
    sigma_Z4(i) = -M_y/(6*A_B*s_z);
    sigma_Z5(i) = M_y/(6*A_B*s_z);
    sigma_Z6(i) = M_y/(3*A_B*s_z);

    P_Z1(i) = sigma_Z1(i)*A_B;
    P_Z2(i) = sigma_Z2(i)*A_B;
    P_Z3(i) = sigma_Z3(i)*A_B;
    P_Z4(i) = sigma_Z4(i)*A_B;
    P_Z5(i) = sigma_Z5(i)*A_B;
    P_Z6(i) = sigma_Z6(i)*A_B;
end

dydz_B2 = 0.4742; % dy/dz for Boom 2 (m)
dxdz_B2 = -0.2738; % dx/dz for Boom 2 (m)

dydz_B3 = 0; % dy/dz for Boom 3 (m)
dxdz_B3 = -0.5476; % dx/dz for Boom 3

sigma_Y2 = sigma_Z2*dydz_B2; % Stress in y-direction of boom 2
sigma_X2 = sigma_Z2*dxdz_B2; % Stress in x-direction of boom 2

sigma_Y3 = sigma_Z3*dydz_B3; % Stress in y-direction of boom 3
sigma_X3 = sigma_Z3*dxdz_B3; % Stress in x-direction of boom 3

for i = 1:1:length(z)
    sigma_2(i) = sqrt(sigma_X2(i)^2 + sigma_Y2(i)^2 + sigma_Z2(i)^2);
    sigma_3(i) = sqrt(sigma_X3(i)^2 + sigma_Y3(i)^2 + sigma_Z3(i)^2);
end

% Calculating shear flows
q = zeros(6,3);
q0 = zeros(6,3);
for i = 1:3
    Dx_w = -1.0751*10^6; % Shear force (N)
    if i == 1
        s_zz = (d-2*z(i)*tand(alpha))/tand(60); % length of side
    end
    if i == 2
        s_zz = (d-2*z(i*10+1)*tand(alpha))/tand(60); % length of side
    end
    if i == 3
        s_zz = (d-2*z(i*10+1)*tand(alpha))/tand(60); % length of side
    end
    Iyy = 3*A_B*s_zz^2; % Moment of inertia (m^4)
    q0(2,i) = q0(1,i) - (Dx_w/Iyy)*A_B*0.5*s_zz;
    q0(3,i) = q0(2,i) - (Dx_w/Iyy)*A_B*s_zz;
    q0(4,i) = q0(3,i) - (Dx_w/Iyy)*A_B*0.5*s_zz;
    q0(5,i) = q0(3,i);
    q0(6,i) = q0(2,i);
    q0_real = -q0(2,i) - q0(3,i) - 0.5*q0(4,i);
```

```
    q(:,i) = q0(:,i) + q0_real;
end
```

Plotting

```
figure(1)
plot(z,abs(sigma_2)*10^-6,'LineWidth',1.5)
hold on
plot(z,abs(sigma_3)*10^-6,'LineWidth',1.5)
hold off
grid on
title('\sigma_{z} Magnitude Along Length of CRV')
xlabel('z (m)')
ylabel('\sigma (MPa)')
legend('Boom 1, 2, 4, and 5','Boom 3 and 6','Location','best')
xlim([0 H])

figure(2)
plot([1,2,3,4,5,6],abs(q(:,1))*10^-3,'LineWidth',1.5)
hold on
plot([1,2,3,4,5,6],abs(q(:,2))*10^-3,'LineWidth',1.5)
plot([1,2,3,4,5,6],abs(q(:,3))*10^-3,'LineWidth',1.5)
hold off
grid on
title('Shear Flow Along Panels of CRV')
xlabel('Panel Number')
ylabel('q (N/mm)')
legend('q_1','q^{bar}','q_2','Location','best')
xlim([1 6])
current_tick = get(gca, 'xTick');
xticks(unique(round(current_tick)));
```